Attention on Lattice Traversal Is All You Need

The EWM Lattice as Map, Attention as Locomotion,
and Materialization as the Door Between Them

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Abstract

Two research programmes have been asking the same question from opposite ends. Transformer *attention* asks: given a query, which stored items should a learned, differentiable routing function read, and how much? The *Emerging World Models* (EWM) lattice asks: given a query *HLLSet*, which content-addressed nodes of a Boolean lattice should an exact, gradient-free routing function visit? We argue that the two are neither rivals nor the same thing: they are the two halves of one journey. **The lattice is the map**—a coarse, exact, persistent, content-addressed graph of places to visit, in which each node is a compact HLLSet that carries the place's full local detail in compressed form. **Attention is the locomotion**—the fine, graduated, differentiable mechanism that actually moves the model between places once their detail has been unpacked. **Materialization is the door**—the morphism that restores an HLLSet context into a token context the LLM can attend to, and, by ingestion, re-compresses the result into a new node. The seam between the two layers is exact and elementary: for nonnegative attention weights, the support of the attention output is the lattice *join* of the attended value nodes, so *the bitmap decides which place is in the working set and the weights decide how the model moves inside it*. The softmax is the entropic (Gibbs) relaxation of a thresholded selection, so the EWM world threshold is the attention threshold and temperature is its smooth relaxation. Around the seam every transformer device has a lattice counterpart: multi-head attention is the family of *worlds*; the output projection is the *bridge*; depth is complement traversal with the Noether stop; causal masking is the *monotone ring basis*; the KV cache is the *commit DAG*; the residual stream is the *union invariant* and the $D/R/N$ decomposition; and Q/K/V projections are the *bootstrap permutation* that closes Ashby's variety gap. The result is a win-win integration: the lattice gives the model a durable, exact, auditable map and a context it can shrink and expand at will; attention gives the lattice a differentiable way to travel, resolve collision detail, and generate the next node. We prove the support–union lemma and the correspondence theorem, state the structural dictionary, describe the travel loop and the exact engine's three modes, and restate "all you need" precisely: attention *on* lattice traversal—the map, the motion, and the door—is the complete primitive.

Keywords: attention; world models; HLLSet; Boolean lattice; content addressing; softmax; traversal; materialization; context management; KV cache; residual stream; IICA.

1 Introduction

Transformer attention [10] is, mechanically, one line:

$$\mathrm{Attn}(X) \;=\; \mathrm{softmax}\!\left(\frac{QK^{\top}}{\sqrt{d}}\right)V, \qquad Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Everything that follows in a modern language model—multi-head splits, causal masks, residual streams, KV caches, flash-attention kernels—is a variation on how that one line is organised,

computed, or fed. The line has been read as retrieval [16], as associative memory [15], as kernel smoothing [13], and as hashing [12]. What all of these readings share is that attention is *content-based routing followed by a monotone accumulation*: a query selects items by a similarity and their values are combined.

The *Emerging World Models* (EWM) programme [1, 2] arrives at the same shape from the opposite direction. Its state is a content-addressed set—an **HLLSet**, a compact bit-vector sketch of a token collection over a fixed 32K-bit plane—whose subsets form a Boolean lattice $\mathcal{B}$. The state machine built on this lattice [3] turns a query into a traversal: a query HLLSet is ingested, a set of stored HLLSet nodes is scored by a symmetric, content-addressed similarity (overlap, containment, BSS), the selected nodes are accumulated by *join* (union), and the result is materialised back to tokens.

The two are close, and the closeness is real—but it is not an identity, and it is not a competition. This paper argues that they are the two halves of one journey.

The traveller's analogy. Before a journey, one consults a *map*. The map names the places worth visiting and how they connect, but it is deliberately coarse: it does not tell you how to board the train. EWM's lattice is that map—a coarse, exact, persistent, content-addressed graph of nodes, each a compact HLLSet. The detail lives *inside* the node: the transportation, the ticket, the hotel are all there, but compressed. The bitmap is the coordinate, not the itinerary. To actually *move* from one place to the next, the traveller must unpack the node's detail and act on it; in our terms, the node is **materialised** into a token context, and the LLM's graduated attention does the moving. The two layers are therefore complementary rather than identical: the lattice answers *which place*, and attention answers *how to travel once there*.

Thesis. *Attention is the locomotion that runs on the EWM lattice's map; the lattice is the exact navigation layer that attention lacks, and attention is the differentiable navigation the lattice cannot provide on its own.* The seam between the two layers is a single exact operator: for nonnegative attention weights, the support of the attention output is the lattice join of the attended value nodes, so the *bitmap* fixes the working set and the *weights* move within it. Materialisation is the door that restores HLLSet context to token context; ingestion is the same door the other way. The integration is a win-win: neither layer subsumes the other, and together they close the travel loop.

Contributions.

1. **The support–union lemma (§4).** For Boolean value nodes V_i and nonnegative attention weights α_i, the support of the attention output is the bitwise join of the attended nodes: $\mathrm{supp}(\sum_i \alpha_i V_i) = \bigcup_{i:\alpha_i>0} V_i$. Hence an attention layer's *bitmap* is a node of the EWM lattice, and the weights live one level up, in its realification. This is the exact seam between the map and the motion.
2. **The correspondence theorem (§4).** A truncated scaled-dot-product attention head is the unique free-energy minimiser of a traversal score, and its support is the θ-thresholded traversal join. The two layers share one support operator and one similarity geometry; they differ only in the convex interpolation inside the support.
3. **The travel loop and the door (§5).** We formalise the node as a place with compressed detail, materialization as the door that unpacks it into token context (with the round-trip composition law as the door's fidelity guarantee), and context management as lattice shrink/expand before materialisation. This is the practical win-win.
4. **The structural dictionary (§6).** Heads are worlds; the output projection is the bridge; depth is complement traversal with the Noether stop; causal masking is the monotone ring basis; positions are rotating coordinates; the KV cache is the commit DAG; the residual stream is the union invariant and $D/R/N$; Q/K/V projections are the bootstrap permutation.

5. **The exact engine (§7).** The three modes—address-key, set-key, and hybrid—form a spectrum from the fully learned relaxation to the exact content-addressed traversal, with the coverage invariant as the exactness gauge.
6. **A precise reading of "all you need" (§8).** The complete primitive is the pair: the map has no motor, the motor has no map, and the door is what lets one hand work to the other.

Roadmap. Section 2 recalls attention. Section 3 recalls the EWM lattice, opening with a one-page HLLSet primer (§3.1) for readers new to the construction. Section 4 proves the seam. Section 5 develops the map-and-journey integration. Section 6 states the structural dictionary. Section 7 describes the exact engine. Section 8 restates the thesis and its limits; Sections 9–11 place it and close.

Remark 1.1 (The Glass Bead Game). Hesse's *Glass Bead Game* [32] is played by threading beads drawn from every discipline into one formal language; the player's skill is the move that makes the whole board legible. This paper is one move. The beads are the objects of the EWM corpus—HLLSets, worlds, the ring, the trajectory, the bridge; the board is the map; and the observation is that the lattice has been drawing the map, attention has been walking it, and materialization has been the door between them all along.

2 Attention, recalled

Let $X \in \mathbb{R}^{n\times d}$ be a sequence of n token states. A single head computes

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \qquad A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \qquad \mathrm{Attn}(X) = AV, \quad (1)$$

with $W_Q, W_K, W_V \in \mathbb{R}^{d\times d_k}$, $A \in \mathbb{R}^{n\times n}$ row- stochastic, and the softmax applied row-wise. Five features make this the mechanism that matters.

1. **Content routing.** The score $\sigma_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j$ is a function of the two states alone. Order enters only through the position encoding and the mask; the routing itself is order-free.
2. **Permutation equivariance.** Permuting the tokens permutes the rows and columns of A and the rows of $\mathrm{Attn}(X)$ identically. Attention cannot tell two orderings apart unless a position signal is added.
3. **Competition.** The row-wise softmax normalises the scores, so keys compete for a fixed unit of weight. This is the selection step.
4. **Value aggregation.** The output is a convex combination of the values, $\mathrm{Attn}(X)_i = \sum_j A_{ij}\mathbf{v}_j$.
5. **Composability.** Heads are concatenated and projected by W_O, and blocks are stacked around a residual stream $\mathbf{x}_{l+1} = \mathbf{x}_l + \mathrm{MHA}_l(\mathbf{x}_l)$ with a normalisation and a feed-forward network.

Two structural remarks are used later. First, the score is invariant under $Q \mapsto QA$, $K \mapsto KA^{-\top}$ for any invertible A [8]; the only invariant object is the matrix of pairwise similarities, and the labelled ontology attached to K is an interpretation, not part of the mechanism. Second, because softmax is strictly positive, a plain attention row has *no zeros*: $A_{ij} > 0$ for every pair. Selection in attention is therefore never hard; it is a matter of degree, and any hard retrieval must be imposed by a threshold or a sparsity mask. Both remarks reappear as the two axes of the seam: *similarity* and *threshold*.

3 The EWM lattice, recalled

We keep this section to the minimum needed for the seam; the full development is in the main EWM paper [1], the lattice/world companion [2], and the state-machine paper [3].



Figure 1: A toy HLLSet plane (sixteen positions, for the picture only). The top two rows sketch two token collections, the bottom two their join and meet. Set operations are bitwise and exact; position 3 is shared because `cat` and `mat` collide there, and the LUT—not the bits—records which token is which.

A primer first. Section 3.1 gives a one-page introduction to HLLSets for readers meeting the construction for the first time; the rest of the section recaps the corpus's notation.

3.1 A short introduction to HLLSets

An **HLLSet** is a compact, fixed-size *sketch* of a collection of items—here, tokens. Instead of storing the items, it stores a bit-vector whose set bits mark the positions the items hashed into. A collection of ten tokens and a collection of ten million become the same kind of object: 32,768 bits, or 4 KB packed. The sketching idea is borrowed from Bloom filters and HyperLogLog [20, 19]—a set represented by a fixed-size pattern of hash-derived positions—but EWM uses it differently: the sketch is not a cardinality estimator, it is the *state*, and all sketches share one plane, so set operations on them are exact bit operations.

How one is built, in one breath. Fix the geometry once. Precision P gives $\mathsf{M} = 2^P$ registers of b bits each, so the **bit plane** $\mathbb{B} = \{0, \ldots, \mathsf{M}\cdot b - 1\}$ has 32,768 positions at the defaults $P = 10$, $b = 32$. Then, for each item t:

1. hash it once, $h(t)$;
2. split the hash into a register index and a trailing-zero state,

$$\mathrm{reg}(t) = h(t) \bmod \mathsf{M}, \qquad \mathrm{tz}(t) = \mathrm{clamp}\big(\mathrm{trailing_zeros}(h(t) \gg P),\, 0,\, b-1\big), \tag{2}$$

   naming a single **bit address** $\mathrm{bit}(t) = \mathrm{reg}(t) \cdot b + \mathrm{tz}(t)$;
3. set that bit—OR it into the sketch.

Nothing else happens. Order does not matter (OR is commutative) and repetition does not matter (OR is idempotent). A side index, the **LUT**, records for each bit which items hashed there, so the process can be read back.

Illustration. Suppose three tokens land as follows (Figure 1; addresses chosen for the picture, on a toy plane of sixteen positions rather than 32,768): `cat`→ 3, `sat`→ 9, `mat`→ 3. Then $H(\{\texttt{cat}, \texttt{sat}\}) = \{3, 9\}$ and $H(\{\texttt{mat}\}) = \{3\}$; the union is $\{3, 9\}$, the meet is $\{3\}$, and the LUT reads bit 3 back as $\{\texttt{cat}, \texttt{mat}\}$. Two different tokens can share a bit—a *collision*—which is why the LUT, not the bitmask, carries the meaning.

What it buys. Three properties are used throughout this paper. *Fixed size*: a waypoint is always 4 KB, so a model can hold a great many of them—this is what makes the lattice a usable map. *Exact set algebra*: union, intersection, difference, and symmetric difference are bitwise and exact on the shared plane, and two sketches are compared by their overlap (Lemma 4.2). *Monotone, content-addressed accumulation*: union is idempotent, commutative, and associative, so independent contributions merge without coordination, and the SHA-1 digest of the bits is the sketch's name. The one price is collisions, and they are managed by the LUT and by re-hashing under further seeds (§7).

From the sketch to the lattice. Write $H(A) = \bigcup_{t \in A} \{\text{bit}(t)\}$ for the sketch of a token collection $A \subseteq \mathcal{T}$. Because every HLLSet is a subset of the one common plane, sketches are automatically a lattice under inclusion:

$$\mathcal{B} = \big(2^{\mathbb{B}},\ \subseteq,\ \cup,\ \cap,\ \Delta\big), \tag{3}$$

with join $\cup$ (bitwise OR), meet $\cap$ (AND), and ring addition Δ (symmetric difference). $2^{\mathbb{B}}$ is a Boolean lattice; equivalently, if we read a bit-vector as a $\{0,1\}$ vector, $\mathcal{B}$ is the vertex set of the 32K-dimensional hypercube. This lattice is the map the rest of the paper traverses.

IICA morphisms. The token-to-node connection is a chosen **measurement morphism** $\mu : \{\text{token collections}\} \to \mathcal{B}$. The model requires only that μ be **IICA**: *Idempotent* (the same collection always yields the same bits), *Immutable* (bits are never mutated; only new nodes appear), and *Content-Addressed* (a node is named by a digest of its content). Any algorithm satisfying IICA is interchangeable; the SHA-1 digest of an HLLSet is its identity, and "existence is registration" [2]. Two instances matter here: the *flat projection* μ_{flat}, which is a join homomorphism $\mu_{\text{flat}}(A \cup B) = \mu_{\text{flat}}(A) \cup \mu_{\text{flat}}(B)$, and *windowed n-gram ingestion* μ_{ng}, which is inclusion-isotonic but not a join homomorphism.

Worlds. The hash function enters as a parameter of the measurement, not of the lattice. Write $w = (h, s, \tau)$ for a hash configuration (h, s)—hash type and seed—together with a BSS threshold τ. A world induces a measurement $\mu_w : 2^{\mathcal{T}} \to \mathcal{B}$, a thresholded node set

$$\mathcal{N}_\tau(w) = \big\{X \in \mathcal{B}_w : \mathsf{BSS}_w(X) \geq \tau\big\}, \qquad \mathsf{BSS}_w(X) = \frac{|X \cap B_w|}{|B_w|}, \tag{4}$$

and an *interpretation* $\iota_w : \mathcal{B} \to 2^{\mathcal{T}}$ that reads a node's bits back through the world's LUT $\mathcal{L}_w$. Worlds share the representation plane and separate only in their interpretation [2]; two worlds communicate only through the **bridge**

$$\Phi_{w_1 \to w_2} = \text{ingest}_{w_2} \circ \text{transcribe} \circ \text{materialize}_{w_1}, \tag{5}$$

which is deterministic, idempotent on builders, and monotone, and which carries *encodings, not content.*

Traversal and the state machine. The EWM roles of the three attention objects are already written down in the cortex design notes [5]: $Q = I(\text{query})$ is a fresh ingestion of the query into the lattice; K is the stored context sub-lattice, content-addressed by CID; $V = M(K \cap Q)$ is the materialised tokens of the matched subset; and the score is a lattice morphism—BSS containment, Jaccard, or overlap popcount. The state machine [3] advances a state by

$$H(t) = \big(S(t),\ H(t{-}1),\ D,\ R,\ N\big), \tag{6}$$

where $S(t)$ is the emerging working state, $H(t{-}1)$ the committed base, and

$$N = S(t) \setminus H(t{-}1), \qquad R = S(t) \cap H(t{-}1), \qquad D = H(t{-}1) \setminus S(t) \tag{7}$$

are the new, retained, and departed parts. The ring keeps a *monotone basis* of the span of the committed originals; the span only grows, coordinates transport exactly across commits by a change-of-basis matrix, and one turn is the six-step transformation *encode, ingest, join, ring-push, commit, materialise* [3]. The context design [9] adds the piece this paper needs most: a conversation is itself an HLLSet, and *materialize-to-prompt* turns that HLLSet context into a conventional LLM prompt.

4 The seam: where attention and traversal meet

4.1 Traversal

Definition 4.1 (Traversal)**.** *Let $\mathcal{B}$ be the EWM lattice and let $\mathcal{B}_{\mathbb{R}} = \mathbb{R}^{\mathbb{B}}$ be its* realification*, with $\mathcal{B} \hookrightarrow \mathcal{B}_{\mathbb{R}}$ the inclusion of a bit-vector as a $\{0,1\}$ vector. A **traversal** of a keyed set $\mathcal{K} = \{(K_i, V_i)\}_{i=1}^{n} \subseteq \mathcal{B} \times \mathcal{B}$ by a query $Q \in \mathcal{B}$ with score $\sigma : \mathcal{B} \times \mathcal{B} \to \mathbb{R}$ consists of a* selection *and an* accumulation*:*

$$\textit{select:}\quad i \in S \iff \sigma(Q, K_i) \textit{ passes a rule}, \qquad \textit{accumulate:}\quad R = \textstyle\bigcup_{i \in S} V_i.$$

*The **soft traversal** readout at temperature $\tau > 0$ replaces the hard selection by the convex aggregation*

$$R^{(\tau)}(Q) = \sum_{i=1}^{n} \alpha_i(\tau)\, V_i, \qquad \alpha_i(\tau) = \frac{e^{\sigma(Q,K_i)/\tau}}{\sum_j e^{\sigma(Q,K_j)/\tau}}, \tag{8}$$

*and the **thresholded traversal** readout at level θ replaces it by*

$$\mathrm{Tr}_\theta(Q) = \bigcup_{i:\, \sigma(Q,K_i) \geq \theta} V_i. \tag{9}$$

*A traversal is **lattice-compatible** when σ factors through the overlap form (§4.2) and the accumulation is the lattice join.*

Comparing (8)–(9) with (1) is already most of the seam: identify Q with a query vector, K_i with a stored key vector, V_i with a value vector, and σ with the dot product. The only difference is that Definition 4.1 permits a *hard* selection, which attention hides inside the softmax. The next two lemmas make the seam exact.

4.2 The overlap form

Lemma 4.2 (Overlap is a bilinear form)**.** *Identify a bit-vector $X \subseteq \mathbb{B}$ with its indicator $x \in \{0,1\}^{\mathbb{B}} \subset \mathbb{R}^{\mathbb{B}}$. Then for all $X, Y \subseteq \mathbb{B}$,*

$$\langle x, y \rangle \;=\; \sum_{b \in \mathbb{B}} x_b y_b \;=\; |X \cap Y|, \tag{10}$$

so the lattice meet is the multiplication of the inner product on $\mathcal{B}_{\mathbb{R}}$. Moreover the similarity measures used by EWM are functions of this form:

$$\mathsf{BSS}_Y(X) = \frac{\langle x, y \rangle}{|Y|}, \qquad \mathrm{Jaccard}(X, Y) = \frac{\langle x, y \rangle}{|X| + |Y| - \langle x, y \rangle}, \qquad \mathrm{contain}(X, Y) = \frac{\langle x, y \rangle}{|Y|}.$$

Proof. Both sides count the coordinates b with $x_b = y_b = 1$, i.e. the bits in $X \cap Y$; the three displayed identities are immediate from that count. □

Lemma 4.2 is the reason a lattice traversal is a geometry and not merely a set operation. The Boolean lattice embedded in $\mathcal{B}_{\mathbb{R}}$ carries the standard inner product, and its meet is a bilinear pairing; the similarity measures EWM uses are the normalised pairings. A learned transformer score $\mathbf{q}_i \cdot \mathbf{k}_j$ is therefore not a foreign object imposed on the lattice: it is the same bilinear form after a learned change of basis and a learned diagonal metric, i.e. after the linear maps W_Q, W_K. The projections are a *re-coordinatisation* of the overlap form; the form itself is intrinsic. This is why the map and the motion can be joined at all: the lattice's notion of "near" and attention's notion of "similar" are the same form.

4.3 The support of attention is a lattice join

Lemma 4.3 (Support–union)**.** *Let $V_1, \dots, V_n \in \{0,1\}^{\mathbb{B}} \subset \mathbb{R}^{\mathbb{B}}$ be Boolean value nodes and let $\alpha_i \geq 0$. Then*

$$\operatorname{supp}\left(\sum_{i=1}^{n} \alpha_i V_i\right) = \bigcup_{i:\, \alpha_i > 0} V_i, \tag{11}$$

where supp *is the set of coordinates with a non-zero entry and the union is bitwise OR. In particular, for the truncated weights $\alpha_i = \alpha_i(\tau) \cdot \mathbf{1}[\sigma(Q, K_i) \geq \theta]$,*

$$\operatorname{supp}\big(R^{(\tau,\theta)}(Q)\big) = \operatorname{Tr}_\theta(Q).$$

Proof. The left-hand side contains b iff $\sum_i \alpha_i V_{i,b} > 0$. Because every $\alpha_i \geq 0$ and $V_{i,b} \in \{0,1\}$, the sum is positive iff some i with $\alpha_i > 0$ has $V_{i,b} = 1$, i.e. iff $b \in \bigcup_{\alpha_i > 0} V_i$. The truncated statement is the same computation after zeroing the sub-threshold weights. □

Lemma 4.3 is the heart of the paper. Read with Definition 4.1 it says:

> *The support of an attention layer is exactly a traversal join: the layer's bitmap is a node of the EWM lattice. The attention weights never change which bits are kept—they only decide the convex coordinates inside the kept support. The bitmap decides which place is in the working set; the weights decide how the model moves inside it.*

Two consequences are worth separating. First, because all weights are nonnegative, attention cannot cancel support: it can only *add* bits, and the accumulated bitmap is monotone in the set of attended keys. This is the lattice's monotone join appearing inside the transformer. Second, the weights are where attention's arithmetic lives, and the support is where its *state* lives; the two levels—realification and lattice—are exactly the two levels EWM's companion note distinguishes (shared representation, separate reading) [2]. Attention is the map that carries the soft level back down to the lattice level by support—which is precisely what the door of §5 does operationally.

4.4 Temperature is the entropic relaxation of the threshold

Theorem 4.4 (Free-energy characterisation)**.** *Fix scores $s_i = \sigma(Q, K_i)$ and a temperature $\tau > 0$. On the simplex $\Delta = \{\alpha \in \mathbb{R}^n_{\geq 0} : \sum_i \alpha_i = 1\}$ the softmax (8) is the unique minimiser of the free energy*

$$F_\tau(\alpha) = -\langle \alpha, s \rangle + \tau \sum_{i=1}^{n} \alpha_i \ln \alpha_i, \tag{12}$$

and the minimiser over the sub-simplex supported on $\{i : s_i \geq \theta\}$ is the truncated softmax. Consequently

$$\lim_{\tau \to 0^+} \alpha_i(\tau) = \begin{cases} 1/|M|, & i \in M = \arg\max_j s_j, \\ 0, & \text{otherwise,} \end{cases} \qquad \lim_{\tau \to \infty} \alpha_i(\tau) = \tfrac{1}{n},$$

and the support at any τ is the θ-superlevel set $\{i : s_i \geq \theta\}$.

Proof. The Lagrangian of (12) with the multiplier λ for $\sum \alpha_i = 1$ is $-s_i + \tau(\ln \alpha_i + 1) - \lambda = 0$, giving $\alpha_i \propto e^{s_i/\tau}$; the truncated version follows by restricting the simplex. The limits are the standard low- and high-temperature limits of the Gibbs distribution; the support statement is the definition of the restricted simplex. □

Theorem 4.4 identifies the two knobs precisely. The EWM world threshold τ of (4) is the attention threshold θ: it is the *hard* (zero-temperature) selection rule. The transformer's softmax temperature is the *entropic* relaxation of that rule—the device that makes the hard superlevel set differentiable by allowing every key a small weight. "Attention" and "traversal" are therefore not two algorithms with a resemblance; they are the two ends of one temperature axis, and the softmax is the annealing schedule between them.

4.5 The correspondence theorem

Theorem 4.5 (Attention–traversal correspondence)**.** *Let $\mathcal{K} = \{(K_i, V_i)\}$ be a keyed set of Boolean lattice nodes, let σ factor through the overlap form, and fix $\tau > 0$ and θ. Then scaled-dot-product attention (1) with truncated softmax at θ and traversal (9) satisfy:*

1. (support) $\mathrm{supp}(\mathrm{Attn}_\theta(Q)) = \mathrm{Tr}_\theta(Q)$*;*
2. (weights) *the attention row is the unique free-energy minimiser on the θ-supported simplex;*
3. (limits) *as $\tau \to 0$ the attention output tends to the value average over the argmax set and Tr_θ to the join of the argmax set; as $\tau \to \infty$ both keep the same support and the output tends to the uniform average within it;*
4. (equivariance) *both operators are equivariant under a simultaneous permutation of the keys: σ and the join are order-independent.*

Hence attention and lattice traversal share one support operator, one similarity geometry, and one accumulation law; they differ only in the convex interpolation inside the support.

Proof. (1) is Lemma 4.3; (2) is Theorem 4.4; (3) combines the limits of Theorem 4.4 with the fact that the join over a shrinking support converges to the join over the limiting support (finite unions commute with limits of finite indicator families); (4) is immediate because $\sigma(Q, K_i)$ depends on K_i alone and $\cup$ is commutative, so relabelling the keys relabels the scores without changing the support or the join. □

The theorem describes the *seam*, not a merger. It says that where the two layers meet—at the support of a selection—they agree on one operator, so a traversal's output can be handed to an attention layer without translation loss, and an attention layer's output can be handed back to the lattice as a node. It does *not* say that the layers do the same job: the traversal decides *which* nodes are in play and maintains them exactly, while attention decides *how* to move within the detail those nodes carry. The next section makes the division of labour explicit.

5 The map and the journey: materialization as the door

The correspondence theorem is the seam. But the two layers do different jobs, and stating the division of labour explicitly is the paper's practical claim.

5.1 The node is a place with its detail compressed inside

An HLLSet is a *coordinate*, not a transcript. The node a traversal selects is a fixed 32K-bit sketch; everything the place contains—lexical detail, structural order, co-occurrence, local statistics—is either folded into those bits or carried alongside them in the world's LUT and adjacency matrix [9]. This is what makes the map cheap: the traveller can hold a great many destinations at once, because each is a node, and nodes are addressed and compared by content. The detail is not lost; it is *compressed*, waiting behind the door.

5.2 Materialization is the door

To act inside a node, the model needs the detail in a form it can compute on: tokens and their embeddings. **Materialization** is the morphism that restores an HLLSet context into a token

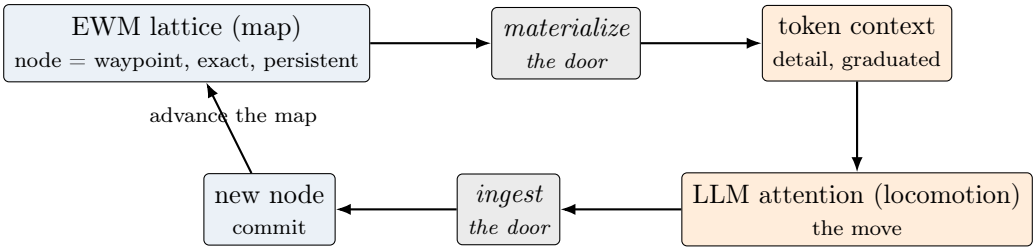


Figure 2: The travel loop. The lattice (map) names the waypoint; materialization (door) restores the node's HLLSet context into token context; attention (locomotion) makes the move inside that detail; ingestion (door, reverse) re-compresses the result into a new node; the commit advances the map. The support–union lemma is the exact seam between the map and the motion.

context—the *materialize-to-prompt* step of the context design [9]:

$$\mathcal{C}_{\text{tok}} \xrightarrow{\text{ingest}} H_{\text{ctx}} \xrightarrow{\text{materialize}} \mathcal{C}'_{\text{tok}}. \tag{13}$$

The door is faithful. The *round-trip composition law* states that when a context passes through the door and back—ingest, then materialize, then ingest again—the 1-HLLSet is recovered exactly, $H_A^{(1)} = H_B^{(1)}$, because the 1-HLLSet is order-independent and materialization emits only tokens whose bits are set in the source. The 2/3-HLLSet structural layers round-trip exactly under a LUT that covers the builders, and with high probability when the LUT is noisy [9]. So the traveller may open the door as often as needed without degrading the map.

5.3 Context management: shrink and expand before crossing

Being inside a node, the model holds a *context*—and, crucially, it can manage that context with lattice operations *before* materialising it:

- **Shrink.** Meet the context with the query ($H_{\text{ctx}} \cap Q$), restrict to a sub-lattice, or take a minimal cover, to focus the token context on the relevant part [9, 3].
- **Expand.** Join with neighbouring nodes, traverse further into the commit DAG, lower the BSS threshold, or add bootstrap views (n-grams) to enlarge the token context [1, 2].
- **Restore.** Materialise the resulting node to token context, and re-ingest the model's output to advance the map [9].

This is the operational meaning of "the node carries its own detail": the context window becomes a *managed working set*, decompressed from the lattice on demand, rather than a linearly growing buffer. The state machine's store/cache/unit separation gives the same picture architecturally: the lattice is the durable, compressed store; the token context is the transient, expanded cache; materialize/ingest moves between them [3]. The map tells the traveller which node to be in; the shrink/expand operations tell the traveller how much of the node to unpack.

5.4 The travel loop

The two layers compose into one loop, shown in Figure 2: the lattice chooses the waypoint; the door unpacks it; attention moves inside it; the result is re-compressed and committed; the map advances.

5.5 The win-win

The table is the argument in miniature. A pure lattice engine can plan exactly but cannot generate; a pure attention model can generate fluently but has no exact, persistent notion of where it is and no bounded-cost way to hold a long history. Joined at the seam, the lattice

	Lattice (the map)	**Attention (the locomotion)**
gives	exact waypoints; durable, auditable memory; controlled shrink/expand; collision groups	differentiable travel; detail resolution; generation of the next step
needs	a motor	a map
horizon	long, discrete, content-addressed	local, continuous, contextual
state	bitmap $S \subseteq \mathbb{B}$ (monotone)	convex coordinates $\mathbf{x} \in \mathbb{R}^d$ (reversible)
failure mode	collision ambiguity at scale	context blow-up; drift; no durable state

Table 1: The win-win. Neither layer subsumes the other: the map has no motor and the motor has no map, and materialization is the door that lets each hand work to the other.

supplies the durability and the address, and attention supplies the smooth motion and the detail. The integration is not a replacement of either paradigm; it is the closure of both.

6 The structural dictionary

The correspondence theorem covers one attention head. A transformer is a *stack* of such heads with several architectural devices around them, and each device has a counterpart in the EWM corpus. This section gives the dictionary (Table 2) and the propositions behind its rows.

6.1 Multi-head attention is a family of worlds

Proposition 6.1 (Heads are worlds)**.** *Let head j use score $\sigma^{(j)}$ and values $V^{(j)}$. If each score factors through an overlap form under a hash/seed configuration $w_j = (h_j, s_j)$, then the multi-head block is the holographic presentation of the world family $\Omega = \{w_j\}$: every head reads the* same *lattice $\mathcal{B}$, and the heads differ only in their interpretation fibres. The output projection W_O is the bridge $\Phi_{\Omega \to w_0}$ of (5) from the head worlds back to the residual world w_0.*

Proof sketch. By construction the head configurations are a finite set of hash configurations with distinct scores. The companion note proves that a family of worlds over one lattice is a fibred set $\mathcal{E}(\Omega) = \biguplus_{w \in \Omega} (\mathcal{N}_\tau(w), \iota_w)$ whose base geometry is the shared lattice and whose fibres are the interpretations [2]. Concatenating head outputs is taking the total space; projecting by W_O is the materialise– transcribe–ingest composition that carries one world's reading into another, i.e. the bridge. Determinism, idempotence on builders, and monotonicity are Theorem 3 of the companion note. □

Proposition 6.1 explains two otherwise puzzling facts. First, the heads *share a representation but not a reading*: they can be edited independently, and the companion note's *virtual isolation* is exactly the head-independence that makes multi-head attention work. Second, the output projection is not a learned mixer over unrelated vectors; it is a *bridge*, and its job is to re-encode the head worlds into the residual world. The empirical finding of the side-car project—"the only model-specific code is a tiny adapter" [2]—is the engineering face of the same statement.

6.2 Depth is complement traversal with a Noether stop

Proposition 6.2 (Depth and the stop criterion)**.** *Let $M_0 \supseteq M_1 \supseteq \cdots$ be the sequence of merged nodes at increasing traversal depth, let C be the accumulated context, and let $\Delta_d = M_d \setminus C$ be the complement of what has already been read. Then $\Delta_0 \supseteq \Delta_1 \supseteq \cdots$ and the traversal terminates exactly when $\Delta_d \approx \varnothing$, which is the Noether convergence condition of the EWM evolution law*

[1]. Attention depth is this stopping problem; a compute-adaptive layer that exits when the complement is empty is the exact analogue of an early-exit attention policy.

Proof. C only grows (join is monotone), so $M_d \setminus C$ only shrinks; a shrinking family of finite sets is eventually constant, and the level at which it first equals the empty set is the first depth at which no new bit can be added. This is the union invariant's conservation statement in the traversal order. □

Proposition 6.2 is the design that the cortex roadmap calls *complement-based temporal attention*: read only the part of each level not already accumulated, quantify it with one overlap computation against the merged node, and stop when nothing new remains [7]. Its cost is one lattice operation per level rather than one per node per level, and its correctness is the union invariant.

6.3 Causal masking is the monotone ring basis

Proposition 6.3 (Causality, positions, and rotating coordinates)**.** *For a causal (prefix) run, let $B(t)$ be the monotone ring basis of the committed originals up to step t. Then masked attention reads only nodes of $\langle B(t)\rangle$, and the change of basis $B(t{-}1) \to B(t)$ transports head coordinates exactly by a GF(2) matrix M: a coordinate vector a at $t{-}1$ becomes aM at t. Positional information is the trajectory frame $\varphi_D(X) = (\mathsf{BSS}(X, D_1), \ldots, \mathsf{BSS}(X, D_m))$; relative position is the rotation of $B(t)$.*

Proof. Causal masking keeps exactly the committed prefix, whose closed span is the ring's monotone span [3]. Monotonicity gives $\langle B(t{-}1)\rangle \subseteq \langle B(t)\rangle$, so every old basis element has unique coordinates in the new basis and transports by the change-of-basis matrix (Proposition 4 of the state-machine paper). The frame statement is the definition of a reading; the ring's per-commit re-pivoting is the rotating coordinate system, measured at 534 re-pivotings over 100 commits on the synthetic clip. □

Causal attention is thus a traversal of the *monotone basis*, and the familiar "attention with positions" is a traversal decorated with a moving frame. The state-machine paper's distinction between a monotone basis and a bounded visibility window is exactly the distinction between a global and a sliding-window attention, with the same trade the transformer makes.

6.4 The KV cache is the commit DAG

Proposition 6.4 (Cache and safe replay)**.** *The key–value cache of an attention layer is a committed, content-addressed stack: a key is the content address of a node, a write is a commit, and the running state is the tip. Because ingestion is idempotent and the join is union, reprocessing an uncommitted turn after a crash yields the same node as processing it once; recovery is a pop, not a rebuild.*

Proof. The cache is a set of IICA nodes; a node's name is its digest, so writing it twice is a no-op and concurrent writes converge by re-registration [2]. The safe-replay principle then gives the stated idempotence [3]. □

This proposition is why a lattice engine does not merely *approximate* the KV cache—it *is* one, with the additional property that the cache is shareable, auditable, and exact. The state-machine paper's storage discipline (store only originals and base HLLSets; keep compounds as provenance-stamped views recomputed on read) is a cache policy for attention keys and values.

6.5 The residual stream is the union invariant

Proposition 6.5 (Residual = accumulation; $D/R/N$ = the write set)**.** *The residual update* $\mathbf{x}_{l+1} = \mathbf{x}_l + \mathrm{Attn}_l(\mathbf{x}_l)$ *corresponds, at the lattice level, to the monotone accumulation* $S(t{+}1) = S(t) \cup H_{t+1}$*; the* $D/R/N$ *decomposition* (7) *is the write/keep/drop reading of one attention step; and the union invariant* $H_{\mathrm{sys}}(t) = \bigcup_i L_i(t)$ *is the conservation law that the residual stream reproduces.*

Proof. By Lemma 4.3 the attention term contributes exactly the join of the attended nodes; adding that join to the running state is the monotone accumulation. The $D/R/N$ identities are (7); that the union of all temporal layers is conserved is the EWM temporal union theorem [1], which is path-independence of the join and hence of the residual stream. □

The EWM reading sharpens an intuition about transformers: the residual stream is not an arbitrary additive channel but a *conserved* accumulation, and the attention layer is the selector that decides which bits enter it. "Learning is redistribution of attention across the lattice, not accumulation of weight updates" [1] is the same statement from the EWM side.

6.6 Q/K/V projections are the bootstrap permutation

Proposition 6.6 (Ashby and the projection family)**.** *The Q/K/V projections and the multi-head split are a bootstrap permutation family in the sense of the Ashby–Bootstrap correspondence: each projection is a view of the same token stream, and the union of views closes the variety gap between the token vocabulary and the lattice plane. The n-gram bootstrap and the projection family are the same construction at different levels of the stack.*

Proof. The Ashby–Bootstrap principle states that mismatched cardinalities are resolved by measurement multiplication rather than capacity reduction [1, 28]. A projection W produces one view XW; a head family produces h views; their concatenation is the bootstrapped measurement $H_{\Pi}(S) = \bigoplus_j H(\pi_j(S))$ with the π_j the projection-induced re-orderings. The n-gram family is the special case where the views are contiguity windows. □

6.7 The dictionary

Transformer	**EWM lattice**
query Q	ingested query HLLSet $I(\text{query})$
key K_i	stored node / content address (`h:`$\langle$sha1$\rangle$)
value V_i	materialised tokens $M(K_i \cap Q)$, or a value node
score $\mathbf{q}\cdot\mathbf{k}$	overlap form $\langle x, y\rangle$ under a learned metric
softmax weights	free-energy minimiser; entropic relaxation of thresholding
attention threshold	world BSS threshold τ
attention output	convex coordinates inside the selected support
supp(output)	traversal join $\mathrm{Tr}_\theta(Q) = \bigcup_{\sigma \geq \theta} V_i$
multi-head	world family Ω; shared lattice, separate LUTs
output projection W_O	bridge $\Phi_{\Omega \to w_0}$
depth L	complement traversal $\Delta_d = M_d \setminus C$
early exit	Noether stop: $\Delta_d \approx \varnothing$
causal mask	monotone ring basis $\langle B(t)\rangle$
position / relative position	trajectory frame; rotating coordinates
KV cache	commit DAG / tip (`ewm-git`)
residual stream $\mathbf{x} \leftarrow \mathbf{x} + \text{Attn}$	union invariant $S \leftarrow S \cup H$
$D/R/N$	departed / retained / new decomposition
Q/K/V projections	bootstrap permutation family (Ashby)
layer norm	renormalisation preserving the invariant
retrieval-augmented generation	Mode B set-key traversal (§7)
flash-attention tiling	a traversal *order* over the key set
node (waypoint)	place with detail compressed inside (HLLSet + LUT + matrix)
materialize-to-prompt	the door: HLLSet context $\to$ token context (§5)
context window	managed working set, unpacked from the node on demand
context shrink / expand	meet with query / cover; union / lower threshold (§5)
continuation hint	lattice-side seed for the next move [9]

Table 2: The dictionary. Each row is a bead of the EWM corpus threaded onto a transformer primitive; the first block is the seam (correspondence theorem), the middle block the architecture (propositions above), and the last block the map–journey interface of §5.

7 The exact engine

The correspondence is not only a reading; it is an implementation. The cortex design notes [5, 6] already specify an attention whose key side is lattice-native, and the three modes there are exactly the points of the temperature axis of §4. In the travel picture, Mode B is the map selecting the waypoint, Mode A is locomotion over the lattice's own key partition, and Mode C is locomotion steered by the map.

7.1 Three modes

Mode A (address-key, tier 1).
A token's key is its bit address, $\mathbf{k}_t = E_{\text{bit}}[\text{bit}(t)]$, with E_{bit} a learned $32{,}768 \times d$ table. Tokens in the same cell share a key; V and context disambiguate within a cell. This is the *relaxed* end: the lattice supplies the partition, learning supplies the metric. The

pure hashing variant $\mathbf{k}_t = \text{one-hot}(\text{bit}(t))W_K$ with W_K fixed is Reformer-style bucketed attention [12].

Mode B (set-key, tier 2/3).
The query and the keys are HLLSets, $Q_{\text{hll}} = I(\text{prefix})$, $K_{\text{hll}} = C$ (the context sub-lattice), the score is lattice containment $\mathsf{BSS}_\tau(Q_{\text{hll}} \subseteq K_{\text{hll}})$ or Jaccard, and the value is $M(K_{\text{hll}} \cap Q_{\text{hll}})$. This is the *exact* end and corresponds to the *map lookup* that picks the waypoint: no dot product over learned vectors, only a lattice morphism and LUT extraction. It is the HLLSet-native attention exercised by CAAL-LLM's `suggest_continuations` routine, and it is the general form of which retrieval-augmented generation [16] is a special case.

Mode C (hybrid).
$\sigma(i,j) = \mathbf{q}_i \cdot \mathbf{k}_j/\sqrt{d} + \lambda\,\varphi(Q_{\text{hll}}, K_{\text{hll}})$, with φ a lattice similarity and λ learned or scheduled. The learned dot-product remains the workhorse; the lattice term steers it toward the selected waypoint, and the contribution is measurable by ablation ($\lambda = 0$ versus $\lambda > 0$).

The three modes are not three algorithms but three temperatures. Mode A is attention with a lattice-shaped key space; Mode B is hard traversal, the map lookup; Mode C is a traversal-steered relaxation. Moving from C to B is exactly lowering the temperature on the lattice term.

7.2 Coverage is the exactness gauge

The exact engine's correctness condition is the *coverage invariant*: the reverse index must cover the active bits of the managed sub-lattice,

$$\mathcal{L}.\text{bits} \;\supseteq\; \text{bits}(C), \qquad \text{confidence}(C) = \frac{\text{resolved bits}}{\text{total bits}} = 1.0. \tag{14}$$

Under coverage, the sub-lattice extraction $M(Q \cap C)$ is deterministic, touches no bit outside C, and is exact up to governing tier: tier 1 exact up to collision groups, tier 2 exact membership by multi-seed quorum, tier 3 disambiguated by 2/3-gram structure and De Bruijn order [5]. The collision statistics are the resolution budget of the key space: with the soldered geometry the pairwise collision probability is $p = 1/3072$, and a sub-vocabulary of n active tokens carries $\binom{n}{2}p$ expected collisions—near-exact for $n \lesssim 300$, and repairable by a second and third seed at $p^2 \approx 10^{-7}$ and $p^3 \approx 10^{-11}$ [5]. This is the price list of the exact end of the temperature axis, and it is the lattice's account of why a small context—one node—can be routed and unpacked exactly.

7.3 Traversal order: tiling and content addressing

Attention is a *set* operation, but it is computed in an *order*, and the order is where the systems work lives. FlashAttention [14] computes exact softmax attention by tiling the key matrix and carrying an online softmax normaliser, so that the operator is realised without materialising the $n \times n$ matrix. In the dictionary this is a traversal order over the key set: the keys are blocks, the running normaliser is the accumulator, and the algorithm is exact because the join and the normaliser are associative.

The lattice engine's traversal order is different in kind. Because keys are content-addressed, the key set is not a contiguous matrix but the commit DAG; traversal follows the DAG from the tip, and the ring basis orders the nodes by their pivots. The order is chosen for *reachability* rather than for memory locality, and the complement stop (§6) prunes it. The two orders are complementary: flash-attention tiles a dense key set for throughput; the lattice traverses a sparse, content-addressed key set for exactness and replay. Both are traversals, and both are exact.

7.4 Traversal signals are measurable

The engine's readings are not hypothetical. On the state machine's synthetic 10×10 scene clip the lattice traversal produces exactly the signals an attention diagnostic would want [3]: the linear residual detects all nine scene cuts (2.9× separation for the windowed morphism), the bit novelty $N = X \setminus \text{cover}$ separates cuts from in-scene drift by 3.75×, symmetric differences lie in the ring span in 394/394 cases while meets and joins do not, and the rotating-coordinate view re-pivots 534 coordinates over 100 commits with the pre-commit residual spiking at every cut. In the dictionary these are the traversal's *novelty*, *surprise*, *representability*, and *re-basing* signals; they are the lattice's account of what a move is doing when it is doing something interesting, and they are the natural triggers for opening the door of §5.

8 Why the pair is all you need

The title is a claim about primitives, and in the two-layer reading it is a claim about a *pair*. "Attention on lattice traversal" names the composite: the map (lattice traversal), the motion (attention), and the door (materialization) that joins them. The claim is that this composite is complete—and that neither half is.

The map has no motor; the motor has no map. This is the whole argument, and Table 1 states it. A lattice engine can decide exactly *where* to go, hold a long history at a bounded cost, and prove that a recomputation is safe; it cannot generate fluent detail or resolve the meaning of a collision without unpacking the node. An attention model can generate fluently and move smoothly through detail; it has no exact, persistent notion of where it is, and its context grows with everything it has seen. Each layer is precisely what the other lacks. The seam (Lemma 4.3, Theorem 4.5) is what makes the join lossless, and the door (the round-trip law, §5) is what makes it safe to cross repeatedly.

Sufficiency of the pair. A composite of the map and the motion supplies each of the five functional properties of attention (Section 2):

Attention property	Supplied by
content routing	lattice traversal: σ is a function of Q, K_i alone (Def. 4.1)
permutation equivariance	σ and the join are order-independent (Thm 4.5(4))
competition	the free energy normalises the weights (Thm 4.4)
value aggregation	the join accumulates values (Lemma 4.3)
composability	IICA composition; the union invariant (Props 6.2–6.5)
detail	materialization: the node's compressed context unpacked to tokens (§5)

The first five rows are the seam; the last row is the job attention does that the map alone cannot, and it is the reason the pair, not the traversal alone, is the primitive.

Necessity. Conversely, any operator that routes by a query–key similarity and aggregates values is an instance of Definition 4.1 with θ equal to its selection rule and τ its temperature (Theorem 4.4); and any bounded-context generator that must persist state across sessions needs the map, because the token context is transient by construction. So the pair is not a choice between two designs; it is the minimal closure of either one.

What the integration buys, concretely.

- **Unbounded, auditable memory at bounded cost.** A node is 4 KB packed; the map holds a long history as content-addressed nodes, and the trajectory is replayable [3, 2].

- **Attention that runs where it matters.** Attention is applied to the selected node's context, not to the whole history; the map prunes the key set before the motion begins.
- **Exactness with a measurable budget.** The waypoint is content-addressed and the door round-trips; collisions are the only inexactness, and they are quantified [5].
- **Controllable context.** Shrink and expand the node with lattice operations before materialising, so the working set is a design choice, not a by-product of sequence length [9].

The seam, once more. In attention, the state is the convex coordinate vector $\mathbf{x} \in \mathbb{R}^d$; selection is soft and reversible, and the support is emergent. In the lattice, the state is the bitmap $S \subseteq \mathbb{B}$; selection is hard and monotone, and the coordinates are the emergent, basis-relative reading. Lemma 4.3 is the bridge between them: support maps one to the other, and the free energy is the interpolation along the way. "Attention on lattice traversal is all you need" therefore has a literal operational meaning: draw the map, walk it with attention, and cross through materialization—the rest is temperature.

9 Related work

Attention. The mechanism is from [10], with the query–key–value reading rooted in additive attention [11]. The retrieval reading is developed by [15], which shows that transformer attention is the update rule of a modern Hopfield network and hence an associative-memory retrieval; our support–union lemma is the lattice-side counterpart of that identification, and it makes the retrieval exact at zero temperature. Efficient attention by hashing and kernels [12, 13] trades exactness for sub-quadratic cost by *changing the key partition*—precisely the K-representation choice that the partition principle of [5] says is free. FlashAttention [14] is the traversal-order result discussed in §7. Retrieval-augmented generation [16] is Mode B, the map lookup.

World models and memory. World models learn a predictive latent state [18]; JEPA-style architectures separate encoder, context, and predictor [17]. EWM differs by making the state *content-addressed* and the transitions *lattice operations*, and the state-machine paper shows the resulting state layer is a side-car that any model can attach [3]; the context design shows the same layer can *generate the prompt* from the stored HLLSet state [9]. The uncertainty-reduction appendix of the main EWM paper [1, 31] reads the traversal's novelty signal as a content-blind primary reinforcer, which is the reward-side analogue of the locomotion story here.

Sketches and content-addressed storage. HLLSets descend from probabilistic cardinality estimation [19] and resemblance sketches [21, 20], but are used here as *state*, not as estimators. Content addressing underlies Merkle trees [23], Git [24], and IPFS/IPLD [25]; the lattice's joins are the CRDT-style merge of [26]. The EWM main paper's Grothendieck-topology target [30] and the companion's geometric-morphism reading of bridges make the world family a site; the correspondence theorem is the finite, explicit stage of that programme.

The originals. The four EWM principles [1], the lattice and its worlds [2], the state machine and its trajectory [3], the HLLSet algebra [4], and the design notes on HLLSet K-space, the LUT transformer, temporal attention, the cortex, and the conversation context [5, 6, 7, 8, 9] are the beads this paper threads. The Boolean-lattice background is [27]; the variety law is [28]; the conservation-law reading is [29].

10 Discussion and limitations

The seam is structural, not numerical. We have shown that the two layers share a support operator and a similarity geometry. We have *not* shown that a given trained transformer's weights equal a given lattice traversal's scores; the correspondence is at the level of the operator, and the learned metric remains learned. The dictionary is an identification of roles, and its value is explanatory and constructive, not an equivalence of trained parameters.

The door is faithful, not lossless in content. The round-trip law recovers the 1-HLLSet exactly and the structural layers up to collision ambiguity [9]. Materialization returns the builders plus any collision neighbours; the door preserves the sketch, not the original text. A traveller who needs the exact prose must keep it as an original node, not as a derived one—which is exactly the state machine's storage discipline.

Soft support is total. Because softmax is strictly positive, an untruncated attention row has full support; Lemma 4.3 then says its bitmap is the join of *all* context values. This is a feature, not a paradox: it is why the residual stream accumulates broadly while the weights carry the motion. The hard traversal is the truncated case, and the truncation is the threshold that EWM makes explicit. An untruncated attention layer is a traversal at positive temperature with $\theta = -\infty$.

Exactness has a resolution budget. The lattice's exact end pays in collisions ($p = 1/3072$ per pair) and in materialisation ambiguity. The engine manages it with a multi-seed quorum, structural layers, and the coverage invariant, but the budget is real; a large active node requires the higher tiers, and the tiers cost computation. The dense end pays instead in quadratic compute and inexact retrieval. The temperature axis is also a cost axis.

The ring is linear; attention is not. The ring's span is a GF(2) subspace, closed under symmetric difference but not under intersection, union, or difference; the meet is the obstruction, and the residual is the exact distance of a node from the context span [3]. Attention's softmax is nonlinear in a different way. The dictionary identifies the ring with the causal prefix, but the two nonlinearities are not the same map, and translating a trajectory reading into a locomotion diagnostic (and back) is an open problem.

Open questions. (1) Does the map/motion reading suggest a controller that opens the door only when a traversal signal fires—a *surprise-gated* materialization policy? (2) Can the exact engine's threshold be made differentiable without losing the bitmap (a soft support with hard bits)? (3) Is there a canonical waypoint-selection policy over the commit DAG analogous to flash-attention's tiling? (4) The companion note's bridges are geometric morphisms only in the order-theoretic sense; the full sheaf condition on the world site is still the EWM programme's open target, and the map/motion reading inherits that openness.

11 Conclusion

The EWM lattice is the map: a coarse, exact, persistent, content-addressed graph of places, each node a compact HLLSet that carries its detail compressed inside. Attention is the locomotion: the fine, graduated, differentiable mechanism that moves the model once a node's detail has been unpacked. Materialization is the door between them: it restores HLLSet context to token context, and ingestion takes the result back to the map. The seam is one lemma—the support of an attention layer is the lattice join of the value nodes it attends to—so the bitmap fixes the working set and the weights make the move; and the softmax is the entropic relaxation of

the traversal threshold, so temperature and threshold are two faces of one knob. Around the seam every transformer device falls into place: heads are worlds, the output projection is the bridge, depth is complement traversal with a Noether stop, causality is the monotone ring basis, positions are rotating coordinates, the KV cache is the commit DAG, the residual stream is the union invariant with its $D/R/N$ reading, and Q/K/V are the bootstrap permutation.

The two paradigms are not rivals and not the same: they are the two halves of one journey, and "attention on lattice traversal" is the complete primitive. The map has no motor, the motor has no map, and the door is what lets each hand work to the other.

The Glass Bead Game is won by the move that makes the board legible. The move here is to see that the beads were never rivals—one set drew the map, another walked it, and the game is the journey.

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A Notation

Symbol	Meaning
$\mathbb{B} = \{0, \ldots, \mathsf{M}b - 1\}$	bit plane (32,768 positions at defaults)
$\mathcal{B} = (2^{\mathbb{B}}, \subseteq, \cup, \cap, \Delta)$	EWM Boolean lattice (the map)
$\mathcal{B}_{\mathbb{R}} = \mathbb{R}^{\mathbb{B}}$	realification; the hypercube's ambient space
$H(A)$	HLLSet of token collection A (a waypoint's coordinate)
μ, μ_w	measurement morphism (IICA)
$w = (h, s, \tau)$	world: hash type, seed, BSS threshold
$\mathcal{L}_w$, ι_w	world LUT and interpretation
$\Phi_{w_1 \to w_2}$	bridge (ingest $\circ$ transcribe $\circ$ materialise)
$\mathsf{BSS}_w(X) = \lvert X \cap B_w \rvert / \lvert B_w \rvert$	BSS against the world basis
$H(t) = (S(t), H(t{-}1), D, R, N)$	state equation (6)
$\mathcal{C}_{\text{tok}}$	token context (the materialised working set)
materialize/ingest	the door between lattice and token context
$\sigma(Q, K)$	traversal score
τ, θ	temperature, threshold
$\mathrm{Tr}_\theta(Q) = \bigcup_{\sigma \geq \theta} V_i$	thresholded traversal join
Attn_θ	truncated attention operator
$\mathrm{supp}(\cdot)$	set of non-zero coordinates
$\langle B \rangle$	GF(2) span of the ring basis B
$\mathrm{res}(X)$	residual of X against the span

B Proofs

Lemma 4.2 (overlap is a bilinear form). Already proved in place. The identities follow by counting b with $x_b y_b = 1$.

Lemma 4.3 (support–union). Already proved in place. The only used facts are $\alpha_i \geq 0$ and $V_{i,b} \in \{0, 1\}$; the result fails if weights may be negative, which is exactly the cancellation that the lattice forbids and the convex stream permits.

Theorem 4.4 (free energy). The Lagrangian computation is in place. Two standard consequences are used: (i) the Gibbs distribution is the unique minimiser because F_τ is strictly convex on the simplex for $\tau > 0$; (ii) the low-temperature limit concentrates on the argmax and the high-temperature limit on the uniform distribution.

Theorem 4.5. (1) Lemma 4.3. (2) Theorem 4.4. (3) The support of the truncated Gibbs distribution is constant in τ for $\tau > 0$ and equal to the θ-superlevel set; as $\tau \to 0$ the untruncated weights concentrate on the argmax, and the corresponding join is the join of the argmax set by finiteness of $\mathcal{K}$. (4) Relabel the keys by a permutation π; then $\sigma(Q, K_{\pi(i)}) = \sigma(Q, K_i)$ as a multiset and $\bigcup_i V_{\pi(i)} = \bigcup_i V_i$ because union is commutative and idempotent.

Proposition 6.1. Invoke the companion note's world-family theorem and bridge theorem; the concatenation is the total space $\mathcal{E}(\Omega)$ and the projection is the bridge composition (5). Monotonicity of the bridge gives that increasing the head set can only add support to the residual world.

Proposition 6.5. Lemma 4.3 gives the support identity; $D/R/N$ is (7); the union invariant is the main paper's temporal union theorem. Path-independence of the join is the reason the residual stream is order-independent, which is the conservation statement.

Round-trip law (§5). The 1-HLLSet direction is exact because materialization emits only tokens whose 1-gram bits are set in the source and ingestion is a union over those tokens, so no bit outside the source 1-HLLSet can be added and no set bit can be missed; the structural layers are exact when the LUT covers the builders and otherwise exact up to collision-induced extra paths that must survive both shape filtering and the 1-/2-gram cross-validation [9].

C Implementation map

Concept	Artifact
bit plane, HLLSet	`hllset_core::HLLSet` (Roaring bitmap)
geometry P, M, b	`hllset_contracts` (P, M, `BITS_PER_REG`)
morphism μ	`hllset_morphisms::ingest / materialize`
working state $S(t)$	`ewm-app::StateCache::working`
base $H(t-1)$	commit tip (`ewm-git`)
$D/R/N$	`ewm-scene::noether` and the commit decomposition
ring basis, push	`ewm-boolring::BoolWindow`
commit DAG, tip	`ewm-git` (`commit`, `head`)
trajectory frame, soft key	`context-tree`, ring projections
world, bridge	hash configuration (h, s); two-pass side-car loop
the door	`hllset-context::to_prompt / restore` (§5)
context shrink / expand	sub-lattice restriction, cover, union, threshold
K-storage interface	`hllset-attn::KStorage` (`key_of`, `candidates`, `confidence`)
Mode A key table	`hllset-attn::bit_table` (E_{bit})
Mode B/C scores	`hllset-attn::setkey / hybrid` (BSS, Jaccard, λ)

Table 3: Concept-to-code correspondence, extending the state-machine paper's map with the attention crates proposed in the cortex design notes [5, 6] and the door of the context design [9].

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